Department of
Computer Science and Engineering
Independent University, Bangladesh

Sec No.	CLOs	Marks Allocated	Marks Obtained	Course Code	CSE200
1	CLO1	40		Course Title	Computer Programming II
2				Term	Spring 2025
3				Test	Mid Term Examination
4				Time	90 Mins
5				Date	March 08, 2025
Total Marks		40		Signature	
Instructions		1. Answer ALL questions. 2. Mobile phones or any electronic gadget is not allowed as a calculator. 3. Partial grading is available.			
Student Name				**Section**	
Student ID					

Section 1: Short Questions (3+2+3)

1. Extend the below code for assessing the electricity bill of at least 10 users with no value return.

```
def calculate_bill(units):
    if units <= 100:
        bill = units * 2
    elif units <= 200:
        bill = (100 * 2) + (units - 100) * 5
    elif units <= 500:
        bill = (100 * 2) + (100 * 5) + (units - 200) * 10
    else:
        bill = (100 * 2) + (100 * 5) + (300 * 10) + (units - 500) * 25
    return bill

units = int(input("Electricity usage: "))
bill_amount = calculate_bill(units)
print(f"Your electricity bill is: {bill_amount} BDT")
```

Write question 1 answer here:

```
def calculate_bill(units):
    if units <= 100:
        bill = units * 2
    elif units <= 200:
        bill = (100 * 2) + (units - 100) * 5
    elif units <= 500:
        bill = (100 * 2) + (100 * 5) + (units - 200) * 10
    else:
        bill = (100 * 2) + (100 * 5) + (300 * 10) + (units - 500) * 25
    print(f"Your electricity bill is: {bill} BDT")

def assess_electricity_bills():
    for i in range(10):
        units = int(input(f"Enter electricity usage for user {i+1}: "))
        calculate_bill(units)

# Call the function to assess bills for 10 users
assess_electricity_bills()
```

2. Write small code to iterate the value of the variable "months"

```
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug']
```

Write question 2 answer here:

```
for day in months
```

3. Write the line number of the following code according to the sequence of execution

```
1   print("Prime number detection")
2   def is_prime(n):
3       if n < 2:
4           return False
5       for i in range(2, n//2 + 1):
6           if n % i == 0:
7               return False
8       return True

9   print("##########")
10  prime_numbers = is_prime(27)
11  print("The result is:", prime_numbers)
```

Write question 3 answer here:

1
9
10
2
3
5
6
5
6
7
11

Section 2: Output Tracing (2+4)

4.

```
a = 73
b = '67'
c = 0
c+=33
c = a + int(b)
print ('Value of c is: ', c+=a)
```

Write question 4 output here:

Error

5.

```
def print_number(x):
    for i in range(x):
        print(i, end =' ')
    return

lines = 5

while lines > 0:
    print_number(lines)
    print()
    lines -= 1
```

Write question 5 output here:

0 1 2 3 4
0 1 2 3
0 1 2
0 1
0

Section 3: Error Tracing (5)

(**Underline the line has error and write the reason for the errors in brief**)

6.

```
def incre_g():
    g_counter
    g_counter += 1
    print( g_counter)
    e_var+=1

def l_variable():
    l_counter = 0
    l_counter += 1
    print( l_counter)
    g_counter += 1

def mixed_sco():
    g_counter
    l_var = 100
    g_counter += 1
    print(g_counter)
    print(l_counter)

def inner_function():
    e_var += 10
    print(enclosing_var)

print(e_var)
incre_g()
print(g_counter)
l_variable()
mixed_sco()
print("counter after mixed_sco call:", l_var)
```

Section 4: Code Refactoring (6)

7. Rewrite the following program using for user defined function with arguments for inputs, loop, and range function. The program must produce the exact output produced by the following program.

```
N = int(input('Number for Multiplication: '))
rows = int(input('Number of Rows: '))

print(f'Multiplication for number {N} upto {rows} rows')

i = 1
while ( i <= rows):
 print(f'{N} x {i} = {N*i}')
 i += 1
```

```
def print_multiplication_table(N, rows):
    print(f'Multiplication for number {N} up to {rows} rows')
    for i in range(1, rows + 1):
        print(f'{N} x {i} = {N * i}')

# Taking user inputs
N = int(input('Number for Multiplication: '))
rows = int(input('Number of Rows: '))

# Calling the function
print_multiplication_table(N, rows)
```

8. Write a function calculates the total fare for an Uber ride based on the distance traveled. The fare is calculated using the following pricing structure:

For the first 5 KM	:	50 BDT per KM + 10 BDT surcharge per KM
For the next 5 KM (6-10 KM)	:	40 BDT per KM + 5 BDT surcharge per KM
For distances between 11-25 KM	:	30 BDT per KM + 2 BDT surcharge per KM
For distances beyond 25 KM	:	25 BDT per KM + 1 BDT surcharge per KM

The function takes the total distance (in KM) as input and returns the total fare in BDT.

9. **Create a list named "results" of and add GPA of 8 students. Total student number must be taken as an input from the user. After inserting all the user information in the list, do the followings:**

    1. Print the full list of GPAs
    2. Sort the list and print the sorted list
    3. Find the max, min GPA in the list
    4. Check how many GPA 3.5 available in the list
    5. Search either GPA 4 in the list or not
    6. Remove the latest element in the list
    7. Find the average GPA of students and print the average value.